LESSON 12 · MISSION GUIDE · THE FINALE

Retire MACRO-9000

Two sessions. One report. Everything you've learned, in one project folder.

From **Gerald Brightwell, Managing Director**
To **Jen, People Analytics Lead**
Subject **The board meeting (Thursday week, 10:00, the good biscuits)**

Jen. Twelve lessons ago I told you I didn't know what a Python was. I still don't, strictly, but I've seen the charts and I've stopped being afraid of Mondays.

Here's the ask. The board wants our gender pay gap, properly done, the way the big firms publish it: the six figures, the quartiles, a picture, and an Excel they can take on the train. And one sentence I can say out loud. MACRO-9000 has been telling us 24.6% since 2019 and nobody has ever been able to explain why.

If your numbers hold up, we switch the macro off on Friday. Priya has bought a small cake. No pressure. (Some pressure.) — G.

What you'll build

```
lessons/12-retire-macro-9000/
├── data/brightwell_payroll.csv     200 employees
└── my_work/
    ├── paygap.py                   your functions (load, gap, bonus, quartiles …)
    ├── check1.py … check5.py       one tiny script per exercise: import paygap, print
    ├── report_console.py           session 2: the report on screen
    ├── report.py                   session 2: the report, a PNG and a workbook
    └── output/                     quartiles.png, pay_gap_report.xlsx
```

What you'll be able to do after today

- [ ] Explain the **six UK pay gap figures** and compute each from a payroll file
- [ ] Structure a real piece of work as a **module of functions** plus small scripts that use it
- [ ] Say why a **median** gap and a **mean** gap differ, and what that difference means
- [ ] Turn a table of numbers into **one sentence** a managing director can say out loud
- [ ] Ship it: a chart, a formatted workbook, a pull request with a title that means something

Why this matters at work

- **This is the actual job.** Pay gap reporting is a legal duty for 250+ employers and a credibility test for everyone else. You'll have done it from raw data to board pack, repeatably, in under 100 lines.
- **Functions are how you stay honest.** One `gap_percent` used everywhere means one place to be right. Dave's macro had the formula typed out eleven times, three of them differently.
- **Next April, it's one command.** New payroll file in `data/`, `python my_work/report.py`, done.

How the two sessions work

SESSION 1
The numbers
15 min briefing, then exercises 1–4: hourly rate, the two pay gaps, the bonus gap, the quartiles

SESSION 2
The report
10 min recap, then exercises 5–7: by department, the console report, chart + workbook

LAST 15 MIN
Dave's Leftovers #12
MACRO-9000's own calculation. Find out why it said 24.6%.

Things that will bite you

- Gaps are a share of **men's** pay. Divide by the wrong group and you get Dave's number.
- Bonus gaps count only people **with** a bonus. "Received a bonus" counts **everyone**. Two different denominators.
- Hourly, not annual. Part-timers have small salaries and normal hourly rates.
- `paygap.py` never prints. If you want to see something, print it in a check script.

You're done when...

- [ ] `python my_work/report.py` prints the report on the sheet and writes the PNG and the workbook
- [ ] Every function in `paygap.py` has a docstring, and no function prints
- [ ] Dave's port agrees with you, and your three commit messages say exactly why it didn't before
- [ ] The pull request is called **Retire MACRO-9000** and contains Gerald's sentence

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After this: there is no lesson 13. There's a Friday, a small cake, and a macro being switched off. Then there's every other spreadsheet at Brightwell, and you know what to do with them now.